BSc (Hons) in Information Technology
Year 3

Lab Exercise – Cognitive Functional Size

SE3010 – SEPQM **Semester 1**

1. Consider the following code snippet and answer the questions below.

```
public class NumberCheck{
 public static void main(String[] args) {
   Scanner sc = new Scanner(System.in);
   System.out.println("Enter a number: ");
   int n = sc.nextInt();
   if (n % 2 == 0){
       System.out.print(n + " is an even number ");
       if (n % 6 == 0) {
          System.out.println("and is divisible by 6");
       }
       else {
          System.out.println("and is not divisible by 6");
       }
   }
   else {
       System.out.println(n + " is an odd number ");
       if(n % 3 == 0) {
           System.out.println("and is divisible by 3");
       }
       else {
           System.out.println("and is not divisible by 3");
       }
   }
 }
}
```

a) Calculate the total Cognitive Weight (Wc).

b) Find the number of inputs and number of outputs.

c) Calculate the Cognitive Functional Size (Sf) of the above code segment.

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2. Consider the following code snippet and answer the following questions below.

```
import java.util.Scanner;
public class PrimeChecker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number to check if it's prime: ");
        int num = scanner.nextInt();
        scanner.close();

        boolean isPrime = true;

        if (num <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i <= num / 2; i++) {
                if (num % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
            System.out.println(num + " is a prime number.");
        } else {
            System.out.println(num + " is not a prime number.");
        }
    }
}
```

Lab Exercise – Cognitive Functional Size

a) Calculate the total Cognitive Weight (Wc).

b) Find the number of inputs and number of outputs.

c) Calculate the Cognitive Functional Size (Sf) of the above code segment.